

Ex.No.4: OFDMA Resource Block Allocation

Date:25-7-26

AIM: To simulate and analyze OFDMA (Orthogonal Frequency Division Multiple Access)

resource block allocation and evaluate system performance in terms of resource utilization and user throughput using MATLAB.

Objective

1. To determine the Number of Resource Blocks (RBs) allocated in an OFDMA system based on available bandwidth and system configuration.
2. To evaluate the Resource Block Utilization (%) by analyzing how efficiently available RBs are assigned to active users.
3. To calculate the User Throughput (Mbps) based on allocated resource blocks and modulation efficiency in the OFDMA system.

Procedure

1. Define the system bandwidth and calculate the number of available resource blocks (RBs) based on RB size (e.g., 180 kHz per RB).
2. Identify the number of active users in the OFDMA system for resource sharing.
3. Allocate resource blocks to each user based on a predefined or equal distribution strategy.
4. Compute Resource Block Utilization (%) using the ratio of allocated RBs to total RBs.
5. Assign modulation efficiency values (e.g., QPSK, 16-QAM, 64-QAM) for each user.
6. Calculate User Throughput (Mbps) using allocated RBs and modulation efficiency.
7. Plot graphs for RB allocation and utilization/throughput analysis using MATLAB.

Objective 1

Matlab Code and Output:

```
clc; clear; close all;  
% System parameters  
bandwidth = 10e6; % 10 MHz  
rb_size = 180e3; % 180 kHz per RB  
% Total Resource Blocks  
total_RB = floor(bandwidth / rb_size);  
% Users  
users = 1:6;
```

```

% RB allocation (simple equal + remainder distribution)
base = floor(total_RB / length(users));
alloc = base * ones(1,length(users));
alloc(1:mod(total_RB,length(users))) =
alloc(1:mod(total_RB,length(users))) + 1;
utilization = (alloc / total_RB) * 100;
disp('Total Resource Blocks:')
disp(total_RB)
disp('RB Allocation per User:')
disp(alloc)
figure
% ----- Graph 1: RB Allocation per User -----
subplot(2,1,1)
stem(users, alloc, 'filled', 'LineWidth', 2)
title('OFDMA Resource Block Allocation per User')
xlabel('User Index')
ylabel('Number of Resource Blocks')
grid on
% ----- Graph 2: RB Utilization -----
subplot(2,1,2)
plot(users, utilization, '-o', 'LineWidth', 2)
title('OFDMA Resource Block Utilization (%)')
xlabel('User Index')
ylabel('Utilization (%)')
grid on

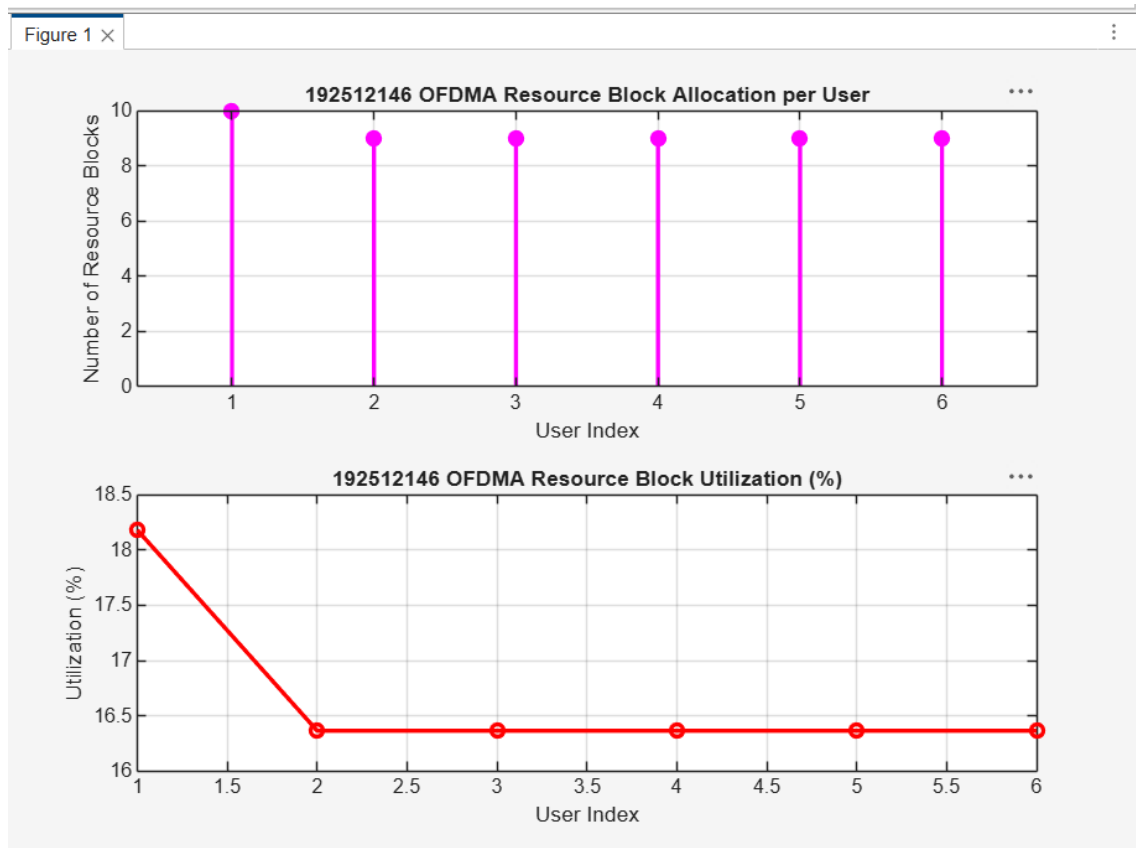
```

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```

/MATLAB Drive/untitled12.m
5  % Total Resource Blocks
6  total_RB = floor(bandwidth / rb_size);
7  % Users
8  users = 1:6;
9  % RB allocation (simple equal + remainder distribution)
10 base = floor(total_RB / length(users));
11 alloc = base * ones(1,length(users));
12 alloc(1:mod(total_RB,length(users))) = alloc(1:mod(total_RB,length(users))) + 1;
13 utilization = (alloc / total_RB) * 100;
14 disp('Total Resource Blocks:')
15 disp(total_RB)
16 disp('RB Allocation per User:')
17 disp(alloc)
18 figure
19 % ----- Graph 1: RB Allocation per User -----
20 subplot(2,1,1)
21 stem(users, alloc, 'filled', 'LineWidth', 2, 'Color', 'magenta')
22 title('192512146 OFDMA Resource Block Allocation per User')
23 xlabel('User Index')
24 ylabel('Number of Resource Blocks')
25 grid on
26 % ----- Graph 2: RB Utilization -----
27 subplot(2,1,2)
28 plot(users, utilization, '-o', 'LineWidth', 2, 'color', 'r')
29 title('192512146 OFDMA Resource Block Utilization (%)')
30 xlabel('User Index')
31 ylabel('Utilization (%)')
32 grid on

```



The first graph shows how available resource blocks are distributed among multiple users in the OFDMA system.

The second graph represents utilization efficiency, indicating how effectively the spectrum is shared among users.

Objective 2

Matlab Code and Output:

```
clc; clear; close all;
```

```
% Total system resource blocks
```

```
total_RB = 50;
```

```
% Number of active users
```

```
users = 1:5;
```

```
% RB allocation to users (example scenario)
```

```
alloc = [12 10 9 8 11];
```

```
% Resource Block Utilization (%)
```

```
util = (alloc / total_RB) * 100;
```

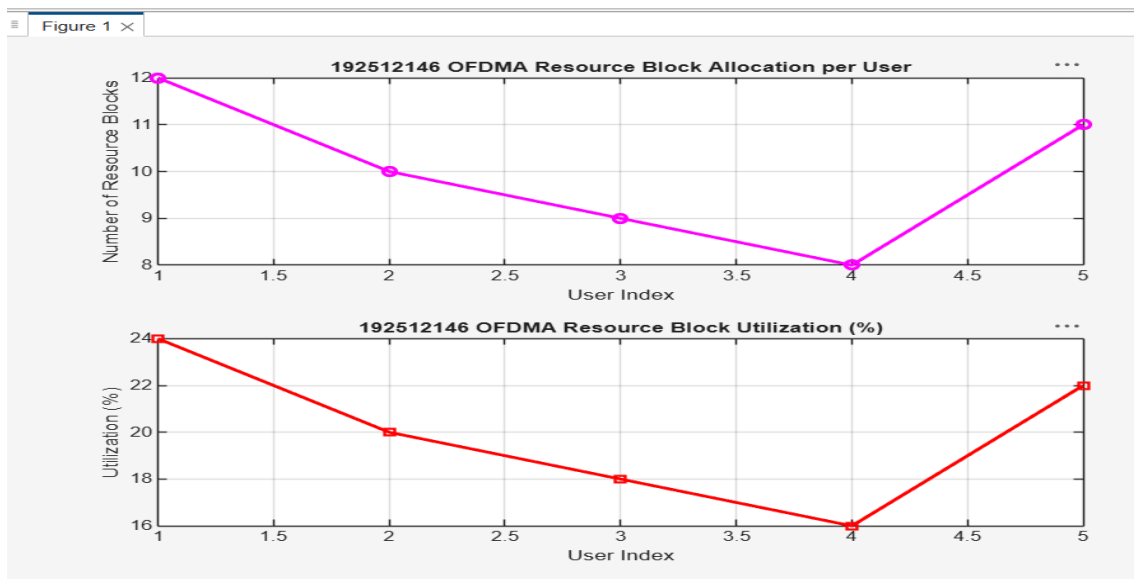
```
disp('Resource Block Utilization (%) per User:')
```

```
disp(util')
```

```
figure
```

```
% ----- Graph 1: RB Allocation -----
subplot(2,1,1)
plot(users, alloc, '-o', 'LineWidth', 2)
title('OFDMA Resource Block Allocation per User')
xlabel('User Index')
ylabel('Number of Resource Blocks')
grid on
% ----- Graph 2: Utilization (%) -----
subplot(2,1,2)
plot(users, util, '-s', 'LineWidth', 2)
title('OFDMA Resource Block Utilization (%)')
xlabel('User Index')
ylabel('Utilization (%)')
grid
```

```
Forward 3.m x exp6.m x exp62.m x exp82.m x exp83.m x exp9.m x untitled12.m x untitled13.m x
e/untitled13.m
1 clc; clear; close all;
2 % Total system resource blocks
3 total_RB = 50;
4 % Number of active users
5 users = 1:5;
6 % RB allocation to users (example scenario)
7 alloc = [12 10 9 8 11];
8 % Resource Block Utilization (%)
9 util = (alloc / total_RB) * 100;
10 disp('Resource Block Utilization (%) per User:')
11 disp(util')
12 figure
13 % ----- Graph 1: RB Allocation -----
14 subplot(2,1,1)
15 plot(users, alloc, '-o', 'LineWidth', 2, 'color', 'magenta')
16 title('192512146 OFDMA Resource Block Allocation per User')
17 xlabel('User Index')
18 ylabel('Number of Resource Blocks')
19 grid on
20 % ----- Graph 2: Utilization (%) -----
21 subplot(2,1,2)
22 plot(users, util, '-s', 'LineWidth', 2, 'color', 'r')
23 title('192512146 OFDMA Resource Block Utilization (%)')
24 xlabel('User Index')
25 ylabel('Utilization (%)')
26 grid on
```



The first graph shows how resource blocks are distributed among multiple users in the OFDMA system.

The second graph indicates the efficiency of spectrum usage by each user in terms of percentage utilization.

Objective 3 Matlab Code and Output:

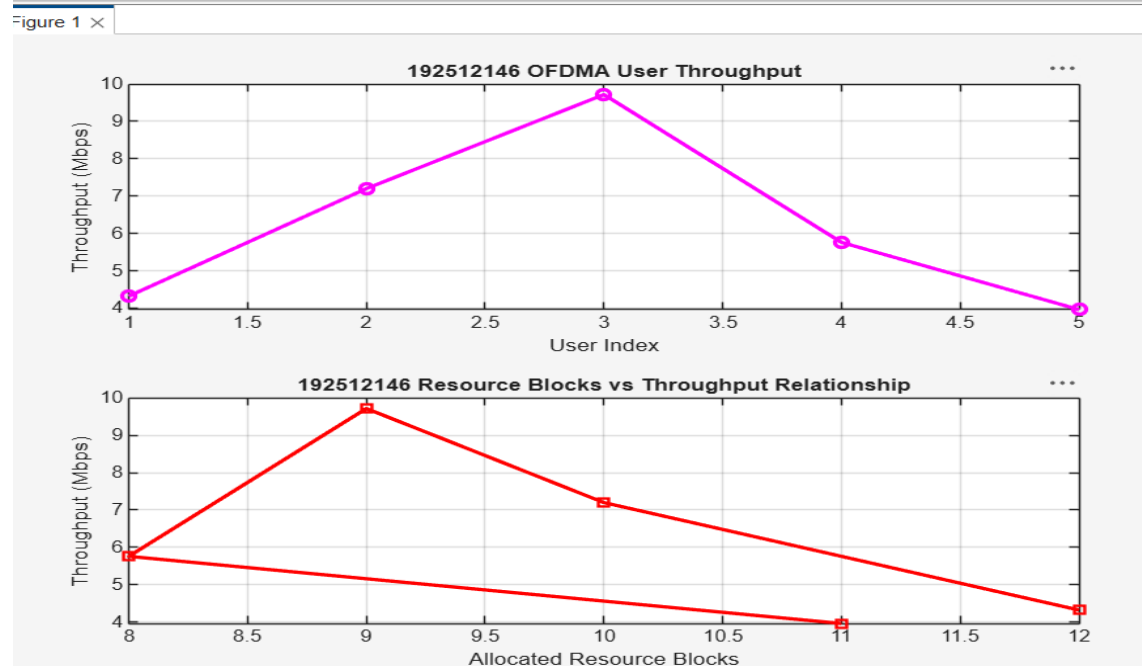
```
clc; clear; close all;
% System parameters
rb_bw = 180e3; % Bandwidth per RB (180 kHz)
users = 1:5;
% Allocated Resource Blocks per user
alloc_RB = [12 10 9 8 11];
% Modulation efficiency (bits/symbol)
mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.
% Throughput calculation (Mbps)
throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
disp('User Throughput (Mbps):')
disp(throughput)
figure
% ----- Graph 1: Throughput per User -----
subplot(2,1,1)

plot(users, throughput, '-o', 'LineWidth', 2)
title('OFDMA User Throughput')
xlabel('User Index')
ylabel('Throughput (Mbps)')
grid on
% ----- Graph 2: RB vs Throughput -----
subplot(2,1,2)
plot(alloc_RB, throughput, '-s', 'LineWidth', 2)
title('Resource Blocks vs Throughput Relationship')
xlabel('Allocated Resource Blocks')
ylabel('Throughput (Mbps)')
grid on
```

```

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IATLAB Drive/untitled14.m
1   clc; clear; close all;
2   % System parameters
3   rb_bw = 180e3; % Bandwidth per RB (180 kHz)
4   users = 1:5;
5   % Allocated Resource Blocks per user
6   alloc_RB = [12 10 9 8 11];
7   % Modulation efficiency (bits/symbol)
8   mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.
9   % Throughput calculation (Mbps)
10  throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
11  disp('User Throughput (Mbps):')
12  disp(throughput)
13  figure
14  % ----- Graph 1: Throughput per User -----
15  subplot(2,1,1)
16
17  plot(users, throughput, '-o', 'LineWidth', 2, 'Color', 'magenta')
18  title('192512146 OFDMA User Throughput')
19  xlabel('User Index')
20  ylabel('Throughput (Mbps)')
21  grid on
22  % ----- Graph 2: RB vs Throughput -----
23  subplot(2,1,2)
24  plot(alloc_RB, throughput, '-s', 'LineWidth', 2, 'Color', 'r')
25  title('192512146 Resource Blocks vs Throughput Relationship')
26  xlabel('Allocated Resource Blocks')
27  ylabel('Throughput (Mbps)')
28  grid on

```



The first graph shows that user throughput increases with higher modulation efficiency and allocated resource blocks.

The second graph clearly indicates a linear relationship between resource blocks and achieved throughput in OFDMA systems.

Result:

1. The Resource Block (RB) allocation was successfully simulated for multiple users in the

OFDMA system, demonstrating how available spectrum is distributed among active users.

2. The Resource Block Utilization (%) analysis showed efficient usage of system resources,